

Robust ecological drought projections for drylands in the 21st century

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Abstract

Dryland ecosystems may be especially vulnerable to expected 21st century increases in temperature and aridity because they are tightly controlled by moisture availability. However, climate impact assessments in drylands are difficult because ecological dynamics are dictated by drought conditions that are difficult to define and complex to estimate from climate conditions alone. In addition, precipitation projections vary substantially among climate models, enhancing variation in overall trajectories for aridity. Here, we constrain this uncertainty by utilizing an ecosystem water balance model to quantify drought conditions with recognized ecological importance, and by identifying changes in ecological drought conditions that are robust among climate models, defined here as when >90% of models agree in the direction of change. Despite limited evidence for robust changes in precipitation, changes in ecological drought are robust over large portions of drylands in the United States and Canada. Our results suggest strong regional differences in long-term drought trajectories, epitomized by chronic drought increases in southern areas, notably the Upper Gila Mountains and South-Central Semi-arid Prairies, and decreases in the north, particularly portions of the Temperate and West-Central Semi-arid Prairies. However, we also found that exposure to hot-dry stress is increasing faster than mean annual temperature over most of these drylands, and those increases are greatest in northern areas. Robust shifts in seasonal drought are most apparent during the cool season; when soil water availability is projected to increase in northern regions and decrease in southern regions. The implications of these robust drought trajectories for ecosystems will vary geographically, and these results provide useful insights about the impact of climate change on these dryland ecosystems. More broadly, this approach of identifying robust changes in ecological drought may be useful for other assessments of climate impacts in drylands and provide a more rigorous foundation for making long-term strategic resource management decisions.

KEYWORDS

global change, grasslands, hot drought, shrublands, soil moisture, water balance

1 | INTRODUCTION

Climate change is substantially altering ecosystems and becoming a recognized challenge for natural resource managers and policy makers (Kemp et al., 2015). Anticipating long-term climate change impacts is particularly urgent in dryland regions, which are characterized by high climatic aridity—low precipitation relative to potential evapotranspiration (PET; Prăvălie, 2016). Dryland ecosystems represent ~40% of the global land surface, directly provide important ecosystem services to >2.5 billion people, and are potentially sensitive to altered climate conditions (Prăvălie, 2016; Reynolds et al., 2007). Globally, many dryland areas are becoming drier (Polade, Pierce, Cayan, Gershunov, & Dettinger, 2014), while others are becoming more mesic (Huang et al., 2016). Over the next century, trends toward increasing temperature, enhanced aridity, and shifts in drought regimes, are all projected for drylands (Dai, 2013; Lin, Gettelman, Fu, & Xu, 2018; Schlaepfer et al., 2017; Trenberth et al., 2014).

Dryland ecosystems are tightly linked to water availability, so future shifts in overall aridity may substantially alter dryland plant communities and the ecosystem services they provide (Bestelmeyer et al., 2015; Reynolds et al., 2007). Vegetation dynamics in dryland regions are driven primarily by availability of soil moisture, which shapes the structure and function of plant communities by affecting photosynthesis, biomass production, plant mortality, and regeneration dynamics (Gremer, Bradford, Munson, & Duniway, 2015; Renne et al., 2019; Schlaepfer et al., 2015; Shriver et al., 2018; Thoma, Munson, & Witwicki, 2018; Witwicki, Munson, & Thoma, 2016). Drought stress can leave ecosystems vulnerable to rapid species turnover and altered structure by reducing plant growth and regeneration. Simultaneously, severe droughts or heat waves may catalyze rapid ecological change in drylands by killing established plants. Ultimately, drought and disturbances may combine in coming decades to promote large-scale transformations in dryland vegetation (Breshears et al., 2005; Martinez Vilalta & Lloret, 2016). Indeed, widespread plant mortality and regeneration failures have already been observed in drylands, and are expected to continue (Breshears et al., 2009; Stevens-Rumann et al., 2017; Williams et al., 2013). Although the strong dependence of dryland ecosystems on moisture may make them especially sensitive to future changes in precipitation and patterns of water cycling (Huang et al., 2017; Huxman et al., 2004), assessing the consequences of climate change for dryland ecosystems is complicated by two primary challenges.

First, long-term climatic aridity does not translate directly into ecologically relevant measures of moisture limitation. Climate change impacts on dryland vegetation are difficult to infer from climate alone, because the drought conditions that most impact vegetation are related to soil moisture deficiency rather than temperature or precipitation (Feng et al., 2018; Gremer et al., 2015). Soil moisture deficiency can be quantified by soil water potential, a measure of how difficult water is to extract from the soil, and plants vary in their tolerance to low water potential conditions (Tucker, Craine, & Nippert, 2011). Ecologically important drought stress, hereafter

ecological drought (Crausbay et al., 2017), can be complex and take various forms (Feng et al., 2018). However, for upland vegetation in drylands, ecological drought is imposed by a combination of low soil water potential and high atmospheric demand for moisture, which is driven by air temperature. Co-occurring high temperatures and low soil moisture exert pronounced stress on plant vascular systems, and these extreme hot-dry conditions have been identified as important drivers of plant mortality (Breshears et al., 2018). Ecological droughts include long-term chronic moisture deficits (Hoover, Duniway, & Belnap, 2017), short-term extreme moisture deficiencies, especially in combination with high temperature (Adams et al., 2017; Williams et al., 2013; Zhou, Zhang, Park Williams, & Gentile, 2019), and moisture deficiencies in particular seasons or soil layers (Sala et al., 1997) that are not easily estimated from meteorological conditions alone (Kulmatiski & Beard, 2013b). Quantifying ecological drought for dryland vegetation is not straightforward because soil moisture varies through time (Silvertown, Araya, & Gowing, 2015) and across depth within the soil profile (Brown & Archer, 1990; Kulmatiski & Beard, 2013a; Sala, Golluscio, Lauenroth, & Soriano, 1989). These temporal patterns are shaped by spatial heterogeneity in edaphic conditions, landscape position, and vegetation structure and function, including plant response to rising atmospheric carbon dioxide (CO₂) levels (Anderegg, Trugman, Bowling, Salvucci, & Tuttle, 2019; Midwood, Boutton, Archer, & Watts, 1998; Milly & Dunne, 2016; Sperry & Hacke, 2002; Swann, Hoffman, Koven, & Randerson, 2016) that further complicate attempts to quantify future dryland ecological drought (Feng, Trnka, Hayes, & Zhang, 2016).

The second challenge for assessing climate change impacts on drylands is uncertainty in long-term precipitation projections. Increasing temperatures and elevated precipitation variability are apparent in historical weather records (Groisman & Knight, 2008; Hartmann et al., 2013) and are consistent features of climate model projections (Collins et al., 2013; Diffenbaugh et al., 2017; Knutti & Sedlacek, 2013; Polade et al., 2014; Prein et al., 2017; Trenberth, Fasullo, & Shepherd, 2015). By contrast, other aspects of climate change do not exhibit clear signals in historical trends or future projections. In particular, changes in precipitation and climatic aridity vary substantially both among climate models and across geographic locations (Burke & Brown, 2008; Collins et al., 2013; Padrón, Gudmundsson, & Seneviratne, 2019). This effective uncertainty in future precipitation confounds efforts to estimate the future severity of drought impacts in support of policy or management.

Despite the challenges of assessing future ecological drought in drylands, focusing on soil moisture represents an opportunity (Seneviratne et al., 2010) because drought metrics based on soil moisture can integrate relatively robust temperature projections with variable precipitation projections (Cook, Smerdon, Seager, & Coats, 2014). Previous studies have examined future soil moisture, but have characterized drought changes in terms of relative changes in average monthly moisture content within coarse resolution global grids that do not represent site-specific soil conditions (Sheffield & Wood, 2008; Wang, 2005). While useful for broad perspectives about future drought, these results may not provide sufficiently

detailed perspectives about ecological drought conditions to guide decision-making at local or regional scales.

To assess long-term climate change impacts in dryland ecosystems, we focused on identifying robust changes in ecologically relevant drought for dryland vegetation. Recognizing the importance of soil moisture, we defined and quantified several drought metrics that represent conditions with demonstrated impacts on dryland plants. Specifically, we examined soil moisture deficiency, in combination with high temperatures, including chronic long-term stress driven by soil moisture deficiency, extreme short-term stress emerging from low soil moisture and high temperature, and seasonal stress imposed by low soil moisture during different times of year. We used an ecosystem water balance model to estimate occurrence patterns of these conditions throughout the 21st century in drylands of the western United States and Canada for all climate models in CMIP5. We addressed three objectives: (a) describe geographic patterns in ecological drought under historical climate, (b) quantify the direction and magnitude of change in ecological drought, (c) identify areas and drought metrics with projected changes that are robust across climate models, defined as locations where >90% of climate models agree in the direction of change.

2 | METHODS

2.1 | Water balance modeling

A complete description of the study design and simulation experiment is provided by Bradford et al. (2019), in which we evaluated ecosystem water balance on a grid with 10 km resolution across the U.S. and Canada and quantified drought metrics in dryland areas where the ratio of mean annual precipitation (MAP) to PET was <0.6. We simulated soil moisture dynamics with the SOILWAT2 ecosystem water balance model (SOILWAT2 v4.2.0, rSOILWAT2 v2.3.2, and rSFSW2 v3.1.2; details available in Github repositories: Schlaepfer & Andrews, 2018; Schlaepfer & Murphy, 2018). SOILWAT2 is a process-based simulation model with multiple vegetation types responsive to atmospheric CO₂ concentrations, multiple soil layers, and a daily time step. The model has been successfully applied in many dryland ecosystems in North America and globally (Bradford & Lauenroth, 2006; Bradford, Schlaepfer, & Lauenroth, 2014; Bradford, Schlaepfer, Lauenroth, & Burke, 2014; Lauenroth & Bradford, 2006; Schlaepfer et al., 2017; Schlaepfer, Lauenroth, & Bradford, 2012; Tietjen et al., 2017). SOILWAT2 is open-source and available as an R package (Schlaepfer & Murphy, 2018). We extracted soil data for all 58,694 grid cells for up to 12 soil layers with a maximal depth of 150 cm. Each soil layer was described with sand %, clay %, volume of gravel, and bulk density from the aggregated databases NRCS STATSGO (1 km² grids; Miller & White, 1998) and ISRIC-WISE v1.2 (5-arcmin; Batjes, 2012) for Canada. We estimated for each grid cell and each simulation run (i.e., depending on the downscaled general circulation model [GCM] inputs and time period) the relative composition of C₃ and C₄ grasses and woody plants as well as monthly

biomass, litter, and root depth distributions from climate conditions using methods described in Bradford, Schlaepfer, Lauenroth, and Burke (2014). We extracted daily weather for the U.S. from Maurer, Wood, Adam, Lettenmaier, and Nijssen (2002) and for Canada from McKenney et al. (2011) for historical climate conditions (1980–2010). For two future periods, 2020–2050 and 2069–2099, we extracted monthly time-series from climate projections (1/8 degree in the United States, 1/2 degree in Canada) from all available GCMs (Table S4.1) for two representative concentration pathways (RCP 4.5–37 GCMs; RCP 8.5–35 GCMs) from the "Downscaled CMIP3 and CMIP5 Climate and Hydrology Projections" BCSO-monthly archive (Maurer, Brekke, Pruitt, & Duffy, 2007) at http://gdo-dcp.ucllnl.org/downscaled_cmip_projections/. We used a hybrid-delta quantile-mapping approach (Hamlet, Salathé, & Carrasco, 2010; Tohver, Hamlet, & Lee, 2014) to obtain future daily forcing from the combination of monthly GCM projections for historical and future time periods with historical daily data. We estimated future atmospheric CO₂ concentrations for both RCP 4.5 and RCP 8.5 with data provided by Meinshausen et al. (2011), downloaded from <http://www.pik-potsdam.de/~mmalt/rccps/> on August 22, 2016.

2.2 | Drought metrics

We examined geographic patterns and future trends in meteorological conditions and ecological drought metrics designed to represent chronic long-term drought, extreme short-term hot drought, and seasonal drought. We examined three meteorological conditions: mean annual temperature (MAT) and MAP and the mean annual ratio of MAP to PET (MAP/PET) that is often used as an aridity index (UNEP, 1992). We quantified chronic drought with four variables: the average annual total number of days with dry soil (<1.5 MPa; a commonly utilized "wilting coefficient" threshold for plants, Brady & Weil, 1999) in shallow (0–20 cm depth) or deeper (>20 cm depth) soil layers, and the length in days of the average longest consecutive dry period in top or bottom soil layers. Chronic drought results were relatively similar between shallow and deep soils, so for the remainder of the manuscript we focus on total dry days in the shallow soil layers that are typically more temporally variable (Choi & Jacobs, 2007) and the length of the longest consecutive period of dry days in the deeper layers that provide important long-term moisture storage (Kulmatiski & Beard, 2013b). We present results for the different depths in the supplementary materials. Recognizing the ecological importance of simultaneous exposure to dry soils and high atmospheric demand for moisture, which is strongly related to air temperature, we quantified extreme drought as the average temperature of the 10 hottest days when soil water potential in all soil layers is less than −1.5 MPa (T_{HOTDRY}). This metric provides a quantitative measure of the stress-inducing gradient between moisture availability and demand that plants must endure during hot drought events. To examine seasonal patterns of ecological drought, we quantified monthly mean soil water availability (SWA) in top and bottom soil layers. SWA was calculated as the amount of water in the

soil greater than the amount at a water potential of -3.9 MPa. This is a useful metric of water availability for studies that encompass diverse edaphic conditions because it provides a consistent measure of availability that is independent of soil texture. We evaluated the number of GCMs that project increases or decreases in each drought metric and define robust projections for changing ecological drought as areas where ecological drought based on SOILWAT2 from $>90\%$ of climate models agree in the direction of change (or $>90\%$ of models agree, for input climate variables like MAT and MAP). Robust drought changes were calculated for each time period and RCP. We used level 2 ecoregions (Omernik, 1987) to explore regional patterns in projected drought trajectories. All analyses on simulation output variables were performed in R version 3.3.2 (R Core Team, 2016).

3 | RESULTS

3.1 | Meteorological conditions

Projected increases in MAT were reasonably consistent across climate models and geographic locations. MAT increases were robust across climate models for 100% of grid cells by both the middle and end of the 21st century under both RCP 4.5 and RCP 8.5 (Table 1; Figure 1). By the end of the 21st century (2070–2100), the mean value across grid cells for the median climate model indicated an increase in MAT of 4.9°C for RCP 8.5 (2.6°C for RCP 4.5). Variation in MAT increases among climate models (10%–90% of models) ranged from 3.5 to 6.4°C (1.6 – 3.7°C for RCP 4.5). By contrast, changes in MAP were much less consistent among climate models and across geographic space. While no grid cells displayed robust signals of decreasing MAP under any timeframe or RCP, increases in MAP were robust under RCP 8.5 for 1.6% of locations by the middle of the 21st century (2.4% for RCP 4.5) and for 20% of locations by the end of the 21st century (2.6% for RCP 4.5; Figure 1). Areas with robust MAP increases occurred in the north central portion of our study

area, in the northern Great Plains and Great Basin. Robust increases in MAP/PET were not observed within our study region, and robust decreases in MAP/PET were not observed for RCP 4.5 under either timeframe or for RCP 8.5 in the middle of the 21st century. Decreases in MAP/PET were robust in 11% of our study area under RCP 8.5, and occur primarily in the southeast portion, in the central and southern Great Plains.

3.2 | Chronic drought

Projected 21st-century trends in chronic drought varied substantially within drylands of the United States and Canada (Figure 2). However, geographic patterns were relatively consistent between our two chronic drought metrics and projected trends were generally similar between shallow and deep soil layers (Appendix S.1). Areas with robust projections for increasing chronic drought included much of the central and southern Great Plains, the southern part of the Colorado Plateau, and the western portion of the Great Basin (Figure 2). By the end of the century under RCP 8.5, 16% of these drylands displayed robust projections for increasing total dry days (7% by mid-century; 9% and 6% under RCP 4.5; Table 1) and 20% displayed robust projections for increasing length of the longest consecutive period with dry deep soils (7% by mid-century; 10% and 7% under RCP 4.5). Robust increases in total dry days ranged from 10 to more than 100, while robust increases in the length of the longest consecutive dry period ranged between 5 and 40 days for the median GCM.

Areas with robust projections for decreasing chronic drought included a large part of the northern Great Plains and Wyoming basins (Figure 2). Under RCP 8.5, 26% of these drylands displayed robust projections for decreasing dry days by the end of the 21st century (18% by mid-century; 22% and 14% for RCP 4.5), while only 9% displayed robust projections for decreasing length of the longest consecutive period with dry deep soils (7% by mid-century; 4% for both time periods under RCP 4.5). Robust decreases in total dry days

TABLE 1 Proportions of North American drylands with robust drought projections, defined as agreement about the direction of change by $>90\%$ of climate models, for RCP 4.5 and RCP 8.5 in the middle and end of the 21st century. + and – indicate robust positive and negative changes, respectively

	2020–2050				2070–2100			
	RCP 4.5		RCP 8.5		RCP 4.5		RCP 8.5	
	–	+	–	+	–	+	–	+
Mean annual temperature	0%	100%	0%	100%	0%	100%	0%	100%
Mean annual precipitation	0.0%	2.4%	0.0%	1.6%	0.0%	2.6%	0.0%	19.9%
Ratio of annual precipitation to potential evapotranspiration	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	12.2%	0.0%
Mean annual total dry days in shallow soils	13.7%	5.9%	17.5%	6.9%	21.5%	8.8%	25.7%	16.5%
Mean annual longest dry period in deep soils	4.2%	6.8%	6.7%	7.0%	4.4%	10.2%	8.8%	20.0%
Mean temperature of 10 hottest days with dry soil	0%	98%	0%	100%	0%	100%	0%	100%

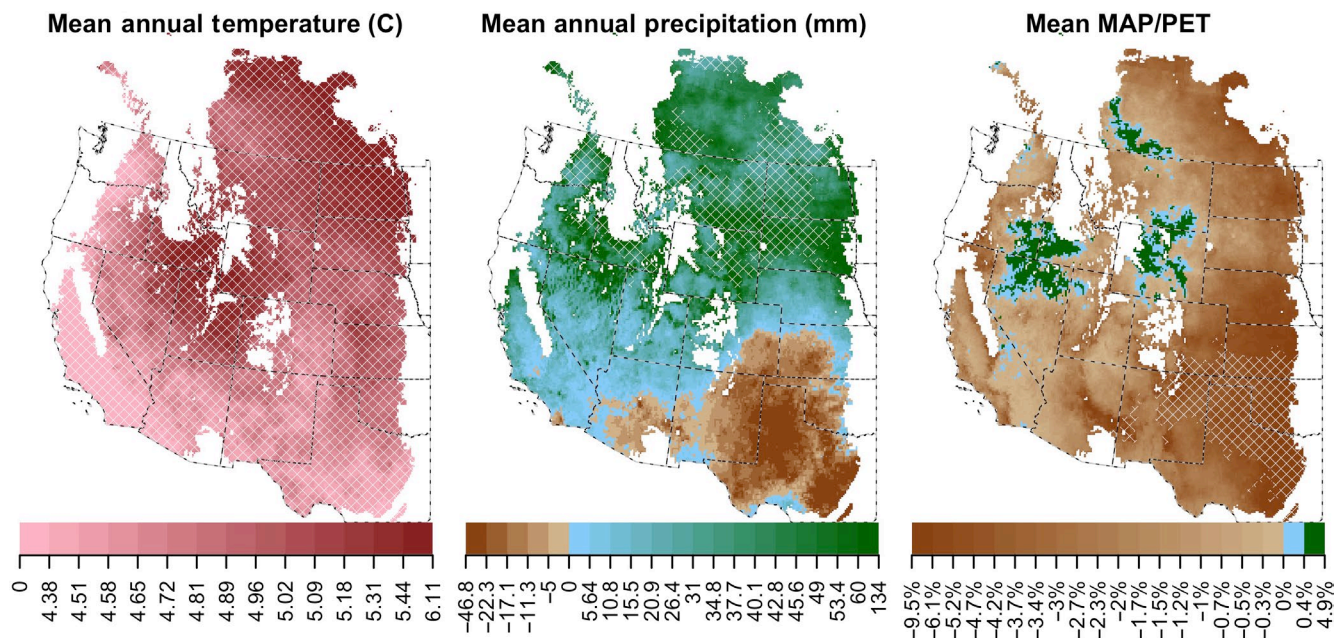


FIGURE 1 Projected change in meteorological conditions. Colors depict average annual change in mean annual temperature (C; left), mean annual precipitation (mm; center), and change in the ratio of precipitation to potential evapotranspiration (precipitation as % of PET, decreasing % indicates that precipitation is lower relative to PET; right) from 1980–2010 to 2070–2100 for the median climate model under RCP 8.5. Cross-hatching indicates areas with robust signals, e.g., where >90% of all climate models running RCP 8.5 projected the same direction of change. Results for RCP 4.5 are presented in the supplementary material

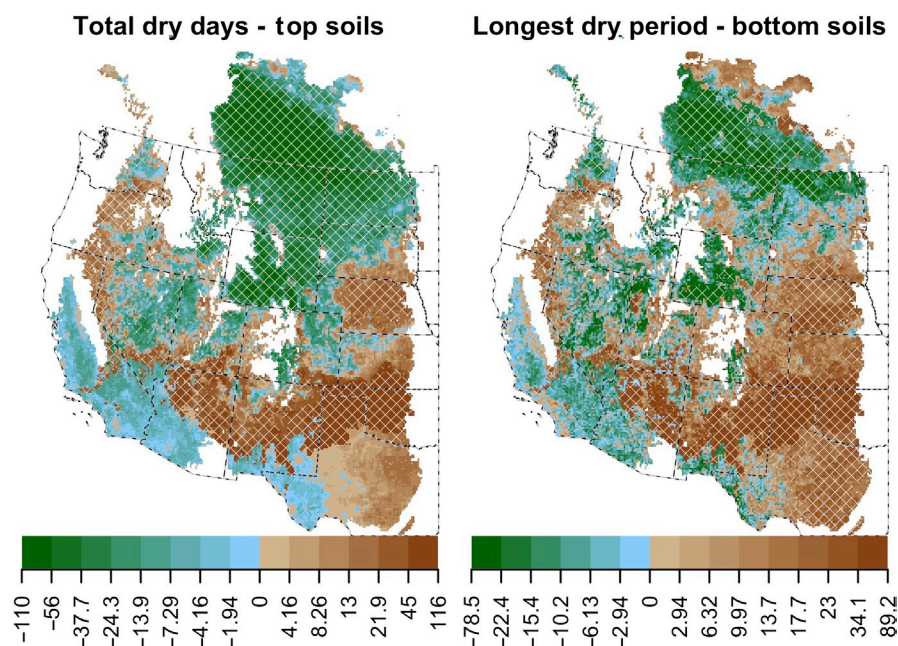


FIGURE 2 Projected change in chronic drought metrics. Colors depict average annual change in the total number of days with dry soil in the shallow layers (left) and average annual length of the longest consecutive period of dry soil in deep layers (right) from 1980–2010 to 2070–2100 for the median climate model under RCP 8.5. Cross-hatching indicates areas with robust signals, e.g., where >90% of all climate models running RCP 8.5 projected the same direction of change. Results for RCP 4.5 are presented in the supplementary material

ranged between 10 and 80, while robust decreases in the longest dry period ranged from 5 to 40 days.

3.3 | Extreme drought

Increasing temperatures during times of low soil moisture displayed robust changes (consistent across climate models) in future drought projections that were similar to expected increases in MAT, but with

some potentially important differences. Hundred percent of our study region (Table 1; Figure 3) displayed robust projections for increasing mean temperature during the hottest 10 days with dry soil (T_{HOTDRY}) under both RCPs by the end of the century and for RCP 8.5 by the middle of the century (98% for RCP 4.5 in the middle of the century; Table 1). By the end of the 21st century, T_{HOTDRY} was projected to increase by an average of 5.1°C under RCP 8.5, and 2.7°C under RCP 4.5 (compared to MAT increases of 5.0 and 2.6°C, respectively). The variation among climate models (10%–90% of models) in T_{HOTDRY} increases

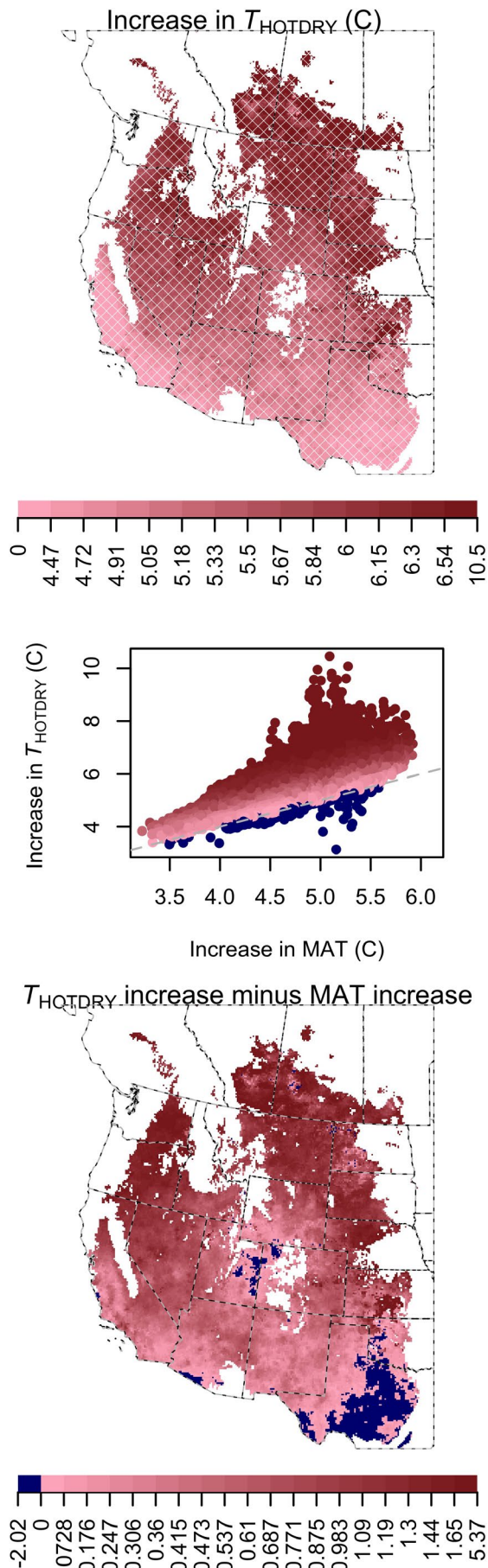


FIGURE 3 Projections for changes in extreme hot drought conditions in North American drylands. Top: by increase in average temperature of the 10 hottest days with dry soils (T_{HOTDRY}), presented for the median climate model under RCP 8.5 in 2070–2100, compared to 1980–2010. Cross-hatching indicates areas with robust signals, e.g., where >90% of all climate models running RCP 8.5 projected the same direction of change. Middle: increase in T_{HOTDRY} versus increase in mean annual temperature (MAT); points are 10 km² grid cells. Bottom: Difference between increase in T_{HOTDRY} and increase in MAT, illustrating more rapid rise in extreme hot drought temperatures than in MATs. Colors for middle and bottom panels both indicate difference between T_{HOTDRY} and MAT

ranged from 3.8 to 6.9°C (1.8–3.8°C for RCP 4.5). Comparing projected increases in T_{HOTDRY} with projected increases in MAT revealed that T_{HOTDRY} was expected to increase more rapidly than MAT for most of these drylands, and particularly in northern areas (Figure 3).

3.4 | Seasonal drought

Seasonal patterns of SWA, and expected 21st-century changes in SWA, varied substantially across these drylands (Appendix S2), so we examined these changes within six ecoregions that represent the broad geographic differences (Figure 4). For all ecoregions, SWA was greatest during the cool season and lowest in the summer and early fall, although the magnitude of average annual peak SWA varied substantially, from ~60 cm in the dryland portions of the Western Cordillera to ~8 cm in the warm deserts. For all ecoregions, SWA changed, either positively or negatively, and the proportion of ecoregions with robust changes were larger under RCP 8.5 than under RCP 4.5, and larger by 2070–2100 than 2020–2050. Most of the robust signals for changing SWA occurred during the cool seasons, while we observed relatively few robust changes during the summer months.

Median projections for changes in SWA suggested increases in the cold deserts and Western Cordillera during the winter months (December–April) that are robust across climate models for a large proportion of these ecoregions. SWA in April and May within these ecoregions varied geographically, with robust decreases in some areas, especially under RCP 8.5 at the end of the century, and robust increases in smaller areas. Summer SWA in these ecoregions was both low and relatively unchanged in the future. SWA in the Temperate Prairies and West-Central Semi-arid Prairies was projected to increase during April and early summer (March–June), and is relatively unchanged during the remainder of the year. In the South-Central Semi-arid Prairies, SWA was projected to decrease during most of the year, with the exception of late summer and fall (June–October). Decreases in SWA were robust in the greatest portion of the South-Central Semi-arid Prairies during fall and winter (November–March). SWA in the Upper Gila Mountains ecoregion generally decreased in spring, early summer (April–June), and early winter (November–December). Projected SWA changes in the Upper Gila Mountains during winter (January–March) were geographically variable, with robust increases in some areas and robust decreases in others. In the Warm Deserts, cool season SWA displayed robust

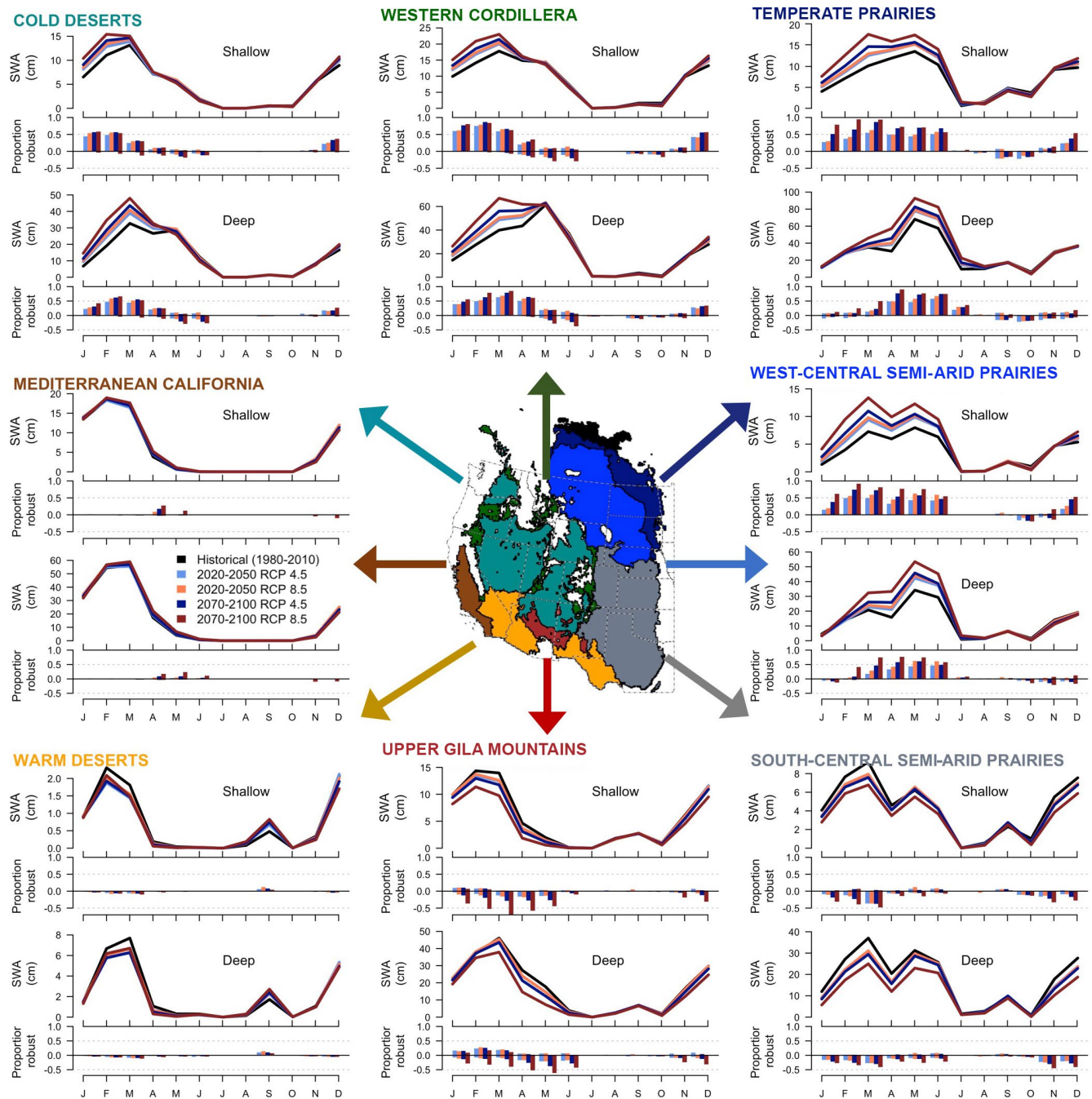


FIGURE 4 Projected future changes in seasonal patterns of shallow (0–20 cm) and deep (>20 cm) soil water availability (SWA) by dryland ecoregion. Within each ecoregion, lines show monthly mean SWA for historic conditions (black) and four future scenarios defined by two time periods (2020–2050 and 2070–2100) and two representative concentration pathways (RCP 4.5 in light blue and blue, RCP 8.5 in orange and red, both for 2020–2050 and 2070–2100, respectively). Bar charts show the proportion of each ecoregion with robust signal (>90% of models agreeing) of positive or negative changes in monthly SWA under the four future scenarios (colors consistent with lines)

decreases over only a small portion of the ecoregion, and SWA changes in Mediterranean California were very slight.

4 | DISCUSSION

Dryland ecosystem structure and function are tightly linked to patterns of moisture availability (Noy-Meir, 1973). Although focusing on

this strong linkage suggests a useful focal point for climate impact studies (Huang et al., 2017), developing insights about long-term ecological responses to climate change in drylands is confounded by two interacting challenges. First, the dynamics of dryland vegetation are dictated by drought conditions that vary in time and space, are difficult to define, and can be complex to estimate from climate conditions alone (Crausbay et al., 2017). Second, high variability in projected climate trajectories, particularly precipitation, generates

important uncertainty about future conditions for water-limited species, communities and ecosystems (IPCC, 2013). Recognizing these challenges, we have focused here on defining ecologically relevant metrics of drought stress, and assessing the robustness of projected changes in those metrics among climate models. Our results provide insight about both the type of drought metrics that display robust signals based on current climate models, and on the geographic patterns of robust drought projections within drylands of the United States and Canada.

Although future precipitation is highly variable among climate models, projections for many drought metrics are more robust (consistent among models) because drought is driven by the combination of temperature and precipitation. As a result, ecological drought metrics that are most directly linked to temperature, notably hot-dry stress, display the largest proportion of robust projections, while metrics more driven by precipitation, notably SWA, have less robust projections. Nevertheless, compared to precipitation, all of our ecological drought metrics displayed robust signals for change over much substantially larger areas. Drought metrics with future projections that are more robust than precipitation includes changes in both chronic drought, seasonal shifts in SWA in many regions, and especially increases in hot-dry stress, which is consistent among climate models throughout these drylands. Projected changes in ecological drought, both increases and decreases, tend to be largest under the most severe climate change and by the end of the century (Figure 4). This positive relationship between the magnitude of climate change and the geographic divergence in dryland drought severity, e.g., wet places becoming wetter while dry places become drier, implies that the spatial heterogeneity of water availability in dryland ecosystems may also increase in the future. Despite these general patterns of ecological drought increasing over time, we did observe some localized examples of drought trends reversing between the middle and end of the 21st century, notably spring SWA in the Western Cordillera and the Upper Gila Mountains, which increases from historical to the 2020–2050 timeframe and then declines in 2070–2100 (Figure 4).

Our results indicate substantial potential for rapidly increasing hot-dry stress, particularly in the northern portions of these drylands. Stressful combinations of high temperature and low soil moisture have been shown to drive mortality of trees (Breshears et al., 2018), and may also apply to other plants (Renne et al., 2019). Our results reinforce other studies anticipating the importance of increasing heat stress for dryland plants (McDowell et al., 2016), and our focus on periods of co-occurring extreme heat and low soil moisture provides new perspectives about patterns in future hot-dry stress. The observation that T_{HOTDRY} increases are, for many areas, projected to be higher than changes in MAT has important implications for estimating potential ecosystem response to future drought. Specifically, more rapid increases in T_{HOTDRY} compared to MAT suggests that utilizing MAT projections to assess conditions during future drought events may underestimate the severity of future ecological drought stress. Furthermore, the magnitude of the difference between projected increases in T_{HOTDRY} versus increases

in MAT is greatest in the northern portion of our study region, areas where precipitation is expected to increase and chronic drought expected to decrease. Parts of the northern Great Plains and northern Great Basin are expected to experience increases in T_{HOTDRY} that are 1–1.5°C higher than their projected increases in MAT. By contrast, T_{HOTDRY} increases are smaller than MAT increases in only small areas, generally in the southern Great Plains, suggesting that those regions may experience less growth in hot-dry stress than would be expected based on increases in MAT alone.

The more rapid increase in T_{HOTDRY} compared to MAT is, at least in part, a result of greater warming during the summer months when soils in many of these northern ecoregions are expected to experience increasingly dry conditions (Appendix S3). Because temperatures are rising particularly fast in the northern portion of the region (Appendix S2), our observation of unexpectedly large increases in hot-dry stress is likely exacerbated by hot spells accelerating soil drying and maximizing the potential for co-occurring hot and dry conditions. Increasing severity of hot droughts may have substantial impacts on the upland plant communities in these areas that are not easily predicted from projected changes in overall aridity. Ecological impacts may include elevated plant mortality (Breshears et al., 2018; Renne et al., 2019) and increasing wildfire severity (Liu, Stanturf, & Goodrick, 2010), thereby increasing bare ground and providing opportunities for invasive species to colonize (i.e., cheatgrass, *Bromus tectorum*), resulting in large potential shifts in species composition. In addition, hot droughts can alter carbon cycling (Zhou et al., 2019). Increasing length of dry spells may interact with increasing exposure to hot-dry conditions. Because the length of a drought required to cause stress is shorter under high temperatures (Adams et al., 2009, 2017), rising temperatures will mean that, in the future, stress may be expected during the shorter droughts that are common in drylands (Lauenroth & Bradford, 2009).

Our results suggest clear differences in future ecological drought among regions within these dryland regions. Depending on the type of ecological drought, different regions appear to have very different long-term trajectories, including potentially decreasing drought stress in some locations and times of year. The largest and most consistent of these differences suggests decreasing chronic drought stress in the northeastern part of our study area (temperate prairies, West-Central Semi-arid Prairies, western Cordillera and portions of the cold desert ecoregions) compared to increasing chronic drought for the southeast part of our study area (South-Central Semi-arid Prairies and upper Gila mountain ecoregions). Our results about increasing aridity in the southern areas are generally consistent with other findings about the US southwest and the southern Great Plains (Cook, Ault, & Smerdon, 2015; Seager et al., 2007; Seager & Vecchi, 2010) but provide additional detail about expected trajectories for different types of ecological drought. Although temperatures in the southern regions are not expected to increase as rapidly as in the north, decreasing precipitation, especially in the southeastern area, combines with higher temperatures to elevate both chronic and extreme drought stress, and to consistently lower SWA during the cool seasons that were

historically relatively wet. Elevated chronic drought (Figure 2) and decreasing cool-season soil moisture (Figure 4) reinforce the potential large impacts of 21st century droughts for dryland vegetation in these regions.

By contrast, in the northern portions of these drylands, where precipitation is expected to increase (Figure 1), many ecological drought metrics suggest increasing moisture availability and presumably decreasing drought stress, while others indicate decreasing moisture and potential for elevated drought stress. In particular, chronic drought displays robust signals for decreasing stress in most of these areas. However, seasonal shifts in moisture availability may have important ecological consequences, especially in combination with elevated hot-dry stress discussed above. Even in these dryland areas where total precipitation is projected to increase in coming decades, warm season drought stress may increase as a result of earlier spring soil drying driven by more rapid evapotranspiration during late winter and spring when water loss was historically slow (Mankin, Seager, Smerdon, Cook, & Williams, 2019; Palmquist, Schlaepfer, Bradford, & Lauenroth, 2016). In addition, elevated storm intensity (Polade, Gershunov, Cayan, Dettinger, & Pierce, 2017) and more precipitation falling as rain (Palmquist et al., 2016) may promote greater water loss to deep drainage in spring. In combination, these effects may alter future seasonal patterns of water balance and moisture availability to include longer, hotter periods of dry soils in summer (Palmquist et al., 2016; Schlaepfer et al., 2017). Specifically, seasonal patterns of soil moisture (Figure 4) suggest that the higher winter precipitation and soil moisture do not persist into the summer months in any ecoregion, including northern areas with clearly elevated winter soil moisture. In addition, other areas with generally decreasing chronic drought stress, including the cold deserts and western Cordillera ecoregions, are projected to experience drier spring soil conditions and drying earlier in the year. These results imply that utilizing the additional moisture resources in these regions may require plants to alter their phenology toward activity in cool seasons. As a result, plants that rely heavily on spring and summer moisture, notably perennial grasses and perennial forbs that provide forage for North American rangelands and comprise the bulk of plant biodiversity (Pennington et al., 2016) may not experience the increased growth potential anticipated by some previous rangeland climate change assessments (Boone, Conant, Sircely, Thornton, & Herrero, 2018; Briske et al., 2015; Polley et al., 2013). In contrast, cool-season plant species that complete much of their growth and reproduction during the late winter to early spring months, such as the invasive grass cheatgrass, may benefit from these shifts in moisture availability with potentially large implications for plant community composition. These more severe dry periods may also lead to decreased transpiration rates, which reduce evaporative cooling and thus increases temperature extremes and the severity of drought impacts on ecosystems (Anderegg et al., 2019).

In addition to this broad latitudinal gradient, our results imply other new perspectives on future trajectories of ecological drought in these drylands. For example, the western Great Basin (e.g., eastern Oregon, northeast CA and southeastern WA), shows signs of

increasing chronic drought stress, specifically robust increases in total dry days. By contrast, other parts of the Great Basin have decreasing, although non-robust, trajectories for chronic drought. Drylands in Mediterranean California are the only ecoregion with modest, non-robust, projected changes in chronic drought and essentially unchanged seasonal patterns of soil moisture availability, although rising temperatures in this region will increase hot-dry stress. In the warm deserts, chronic drought displays small, albeit uncertain, decreases in response to potential increases in late-summer precipitation (Appendix S2) and shows no robust shifts in soil seasonal soil moisture. Elevational gradients are substantial in many of the western ecoregions of the intermountain west, and changes in ecological drought vary across those gradients, although the relationship between elevation and changes in drought differs among the regions. For example, in the Cold Deserts, decreases in chronic and extreme drought are most pronounced at higher elevations (Figure S3.1). By contrast, in drylands of the Warm Deserts, Mediterranean California, Upper Gila Mountains and Western Cordillera, those intermediate or higher elevation areas are projected to have the most substantial increases in chronic and extreme drought stress.

Variation among drought metrics in the projected 21st century changes and robustness highlights the complexity of drought dynamics in drylands. Our long-term drought trajectories indicate that some types of drought stress will become more severe, while other types of drought stress may moderate. Contrasting projections for chronic drought versus extreme hot-dry drought in northern regions is a clear example of this complexity, although our results also suggest potential differences even among closely related metrics. In particular, the two chronic drought metrics that we examined here represent relatively similar types of stress (total number of dry days, and length of the longest dry period), but yielded divergent projections for many locations across North American drylands. Because drought can impact dryland ecosystems in diverse ways (Crausbay et al., 2017), these variable projections underscore the need for a multi-faceted perspective (Zang et al., 2019) on long-term future drought trajectories and potential ecological impacts. This complexity also suggests opportunities for future studies to identify the detailed characteristics of drought that most influence a particular dryland ecosystem, and then focus climate change impact assessments on those specific characteristics. Our projections for robust, rapid increases in hot-dry stress, which may increase more rapidly than anticipated in northern regions, highlight the value of considering complex drought metrics that explicitly represent interactions among temperature and precipitation.

Predictions about long-term ecosystem responses to climate change are complicated by numerous sources of uncertainty (Bonan & Doney, 2018; Mouquet et al., 2015). We have attempted to partially constrain two of these sources: uncertainty about how changing temperature and precipitation will alter conditions that most influence ecosystem dynamics and uncertainty in climate projections. To do so, we have examined ecologically relevant drought conditions with demonstrated influence on ecosystem services, and quantified consistency in future trajectories of those conditions.

We considered a diverse set of drought metrics with recognized importance in dryland environments, and we evaluated all climate models included in CMIP5 under RCP 4.5 and RCP 8.5 to provide a comprehensive assessment of robustness (consistencies across climate models) in projected drought changes. Although variability among climate models projections is an incomplete estimate of actual uncertainty in future climate conditions (Mote, Brekke, Duffy, & Maurer, 2011), this approach is more rigorous than basing ecological decisions on projections of meteorological metrics from one or a few models (Her et al., 2019; Knutti, Masson, & Gettelman, 2013; Knutti & Sedlacek, 2013). This overall approach can be improved by refining the environmental conditions examined, perhaps by synthesizing results from manipulative experiments (Fraser et al., 2012), and/or by maintaining and utilizing monitoring networks. Insights may also be strengthened by integrating these detailed insights about important drought metrics and expected dryland system dynamics into earth system models (Bonan & Doney, 2018).

Our modeling and analysis approach has strengths and limitations. These results are derived from an ecosystem water balance model that represents long-term vegetation patterns instead of short-term dynamic vegetation response to drought. Specifically, while vegetation structure and phenology in the model is influenced by long-term shifts in climate and atmospheric CO₂ concentrations among the historical and future time periods, vegetation structure does not respond to short-term drought or pluvial periods. Reduction of plant leaf area or biomass during droughts may decrease transpiration and moderate the severity of soil moisture deficiency. While it is difficult to estimate the potential magnitude of this biological drought moderation, it will impact both historical and future drought quantification, so is unlikely to substantially influence our assessment of long-term shifts in drought severity. Our modeling approach is also limited to estimating components of water balance and water availability, and does not attempt to quantify processes like plant productivity, mortality or regeneration. Drought metrics from ecosystem water balance models have proven useful in understanding both spatial (e.g., Schlaepfer et al., 2012) and temporal (e.g., Gremer et al., 2015) patterns in dryland plant dynamics, and strengthening these comparisons with future research can help identify the most useful ecological drought metrics (Bradford et al., 2019). A final limitation to this analysis is our focus on average drought over long time periods. Our metrics included some that quantified extreme, short-term drought stress, and seasonal drought stress, but we averaged these values over multi-decadal periods, rather than focusing on, for example, the most severe year. Such episodic extreme drought events can have important impacts on ecosystems and quantifying their trajectories under climate change is both important and challenging (Jentsch & Beierkuhnlein, 2008).

Despite the limitations of the metrics evaluated here, there are some clear implications for long-term planning and management of natural resources in these dryland regions. Recognizing how overall drought conditions are expected to change may improve perspectives about long-term suitability of drought-tolerant dryland plants (Schlaepfer et al., 2012). For example, the Texas-Oklahoma

panhandle region is expected to experience substantial increases in chronic drought. Projections for total dry soil days are robust across models and suggest that by the end of the 21st century, dry days in this region will be more similar to historical conditions in SW Texas (Appendix S1). Results such as these from an array of relevant drought metrics may be useful for designing climate adaptation resource management plans that need to identify plant species or populations with strong potential to maintain viability for decades to come.

These drought projections, particularly the seasonal and depth patterns of changes in SWA, provide perspective about the growing challenge of ecological restoration in these dryland regions. Restoration of plant communities after disturbance or prolonged degradation is an escalating natural resource objective that is particularly difficult in drylands where restoration efforts often fail due to drought-driven mortality (Call & Roundy, 1991; Kildisheva, Erickson, Merritt, & Dixon, 2016). Decreasing spring-time soil moisture in the southern areas suggests elevated potential for seedling mortality. While northern areas are expected to receive more overall precipitation and have wetter soils during the cool season, summer soil moisture availability in the northern ecoregions is not expected to increase (Figure 4). Thus, the elevated moisture during the cool season does not translate into drought mitigation during warm months when plants are actively growing, and indicates that drought-driven mortality during plant establishment may occur even in the wetter northern regions.

Ecological drought projections represent an opportunity to improve assessments of dryland vulnerability to global change. Interactions among drought, disturbances and biological invasions have caused pervasive land degradation in dryland regions (Millennium Ecosystem Assessment, 2005). Estimated geographic patterns in ecological resistance and resilience are increasingly being used to guide the application of land treatments designed to prevent further impacts and rehabilitate degraded areas (Chambers et al., 2017), although current tools are often not built upon drought metrics with demonstrated importance for dryland ecosystems (Bradford et al., 2019). Quantitative estimates of ecological drought severity, like those presented here, are derived from process-based analytical tools that can be leveraged to provide ecologically relevant foundational information about current and future drought patterns to improve these vulnerability assessments.

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DATA AVAILABILITY STATEMENT

Data generated during this study have been released in Bradford and Schlaepfer (2020) and are available from the USGS ScienceBase-Catalog at <https://doi.org/10.5066/P9YU6PQB>.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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